

Omnichannel Retail Sales & Inventory Analytics Dashboard

Project Report

Prepared By

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1. Executive Summary

The Omnichannel Retail Sales & Inventory Analytics Dashboard project was developed to analyze retail transaction data and generate actionable business insights. The project uses Python for data cleaning, MySQL for database management, SQL for analytical querying, and Power BI for interactive dashboard creation.

The primary objective is to transform raw retail sales data into meaningful business intelligence that supports decision-making related to sales performance, inventory management, customer behavior, and revenue analysis.

2. Business Problem

Retail organizations collect large amounts of transactional data from various channels. Without proper analysis, it becomes difficult to:

- Monitor sales performance
- Track revenue growth
- Identify top-selling products
- Analyze customer purchasing patterns
- Monitor inventory levels
- Understand payment preferences

This project addresses these challenges through data analytics and visualization.

3. Project Objectives

The main objectives of the project are:

- Clean and preprocess retail sales data
 - Store data in a structured MySQL database
 - Perform business analysis using SQL
 - Create interactive dashboards in Power BI
 - Generate business insights from retail transactions
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4. Technology Stack

Technology	Purpose
Python	Data Cleaning
Pandas	Data Manipulation
MySQL	Database Management
MySQL Workbench	Database Administration
SQL	Data Analysis
Power BI	Dashboard Development
Git	Version Control
GitHub	Repository Management
VS Code	Development Environment

5. Dataset Description

The dataset contains retail transaction records with the following columns:

Column Name	Description
order_id	Unique Order Identifier
customer_id	Unique Customer Identifier
customer_name	Customer Name
product	Product Purchased
category	Product Category
quantity	Quantity Sold
price	Product Price
order_date	Date of Order
city	Customer City
payment_method	Payment Method Used
inventory_status	Product Inventory Status

6. Project Workflow

The project follows the below workflow:

Raw Dataset → Data Cleaning → MySQL Database → SQL Analysis → Power BI Dashboard → Business Insights

7. Data Cleaning Process

Data cleaning was performed using Python and Pandas.

Tasks Performed

Duplicate Removal

Duplicate records were identified and removed to improve data quality.

Missing Value Handling

Missing values were detected and appropriately handled.

Data Type Validation

Columns were validated and converted into suitable formats.

Date Standardization

Order dates were standardized for accurate monthly analysis.

Data Export

The cleaned dataset was exported as:

cleaned_sales.csv

8. Database Implementation

A MySQL database named:

retail_analytics

was created.

The sales table was designed to store retail transaction records.

The cleaned CSV file was imported into MySQL using MySQL Workbench.

9. SQL Analysis

Several SQL queries were executed to answer business questions.

Total Revenue

Purpose:

Calculate overall revenue generated.

Revenue by City

Purpose:
Identify cities generating maximum revenue.

Top Selling Products

Purpose:
Determine best-selling products.

Sales by Category

Purpose:
Analyze category-wise performance.

Monthly Sales Trend

Purpose:
Track sales performance over time.

Payment Method Analysis

Purpose:
Understand customer payment preferences.

Inventory Status Analysis

Purpose:
Monitor stock availability.

Top Customers

Purpose:
Identify highest spending customers.

10. Power BI Dashboard Development

An interactive dashboard was developed in Power BI.

Dashboard Components

KPI Card

- Total Revenue

Bar Chart

- Revenue by City

Pie Chart

- Sales by Category

Line Chart

- Monthly Sales Trend

Table

- Top Products

Donut Chart

- Payment Method Analysis

Column Chart

- Inventory Status Analysis

Slicers

- City
 - Category
 - Payment Method
 - Inventory Status
 - Month
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11. Business Insights

The dashboard enables users to:

- Monitor overall business revenue
 - Identify top-performing cities
 - Track product performance
 - Analyze category contribution
 - Understand customer payment preferences
 - Monitor inventory distribution
 - Evaluate monthly sales trends
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12. Challenges Faced

Data Quality Issues

Challenge:

Missing values and inconsistent records.

Solution:

Data cleaning using Python and Pandas.

Date Formatting

Challenge:

Different date formats.

Solution:
Standardized date conversion.

Dashboard Design

Challenge:
Presenting multiple business metrics effectively.

Solution:
Interactive visualizations and slicers.

13. Skills Demonstrated

Data Analytics

- Data Cleaning
- Data Transformation
- Data Validation

Database Management

- MySQL
- SQL Query Writing

Business Intelligence

- Power BI Dashboard Development
- KPI Reporting

Software Development

- Python
 - Git
 - GitHub
 - VS Code
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14. Future Enhancements

Potential improvements include:

- Sales Forecasting
- Customer Segmentation
- Inventory Demand Prediction
- Automated ETL Pipelines
- Real-Time Dashboard Integration

15. Conclusion

The Omnichannel Retail Sales & Inventory Analytics Dashboard successfully demonstrates an end-to-end data analytics workflow. The project converts raw retail data into meaningful business insights using Python, MySQL, SQL, and Power BI. It highlights practical skills in data preparation, database management, business analysis, and dashboard development.

The solution can help retail organizations make informed decisions regarding sales optimization, inventory management, and customer engagement.

References

- [Python Documentation](#)
- [Pandas Documentation](#)
- [MySQL Documentation](#)
- [Power BI Documentation](#)
- [GitHub Documentation](#)